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Authors:	Q. Ma, Ed. <i>Huawei</i>	Q. Wu <i>Huawei</i>	M. Boucadair, Ed. <i>Orange</i>	D. King <i>Lancaster University</i>

RFC 10065

A YANG Data Model and RADIUS Extension for Policy-Based Network Access Control

Abstract

This document defines a YANG data model for policy-based network access control, which enables enforcement of network access control policies based on group identity. This YANG data model extends Access Control Lists (ACLs) with date and time parameters to support schedule-aware policy enforcement.

Specifically in scenarios where network access is triggered by user authentication, this document defines a mechanism that eases the maintenance of the mapping between a user group identifier and a set of packet header fields to enforce policy-based network access control. Moreover, this document defines a Remote Authentication Dial-in User Service (RADIUS) attribute that is used to communicate the user group identifier as part of identification and authorization information.

Status of This Memo

This is an Internet Standards Track document.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Further information on Internet Standards is available in Section 2 of RFC 7841.

Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at <https://www.rfc-editor.org/info/rfc10065>.

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1. Introduction

With the increased adoption of remote access technologies (e.g., Virtual Private Networks (VPNs) and Bring Your Own Device (BYOD) policies), enterprises adopted more flexibility related to how, where, and when employees work and collaborate. However, more flexibility comes with increased risks. Enabling office flexibility (e.g., mobility across many access locations) introduces a set of challenges for large-scale enterprises compared to conventional network access management approaches. Examples of such challenges are listed below:

- Endpoints do not have stable and unique IP addresses. For example, Wireless LAN (WLAN) and VPN clients, as well as back-end servers based on Virtual Machines (VMs), can move; their IP addresses could change as a result. Furthermore, mechanisms such as IPv6 temporary addresses [RFC8981] and Network Address Port Translation (NAPT) [RFC3022] may further contribute to address instability and non-uniqueness. This complicates the consistent and efficient access control policy enforcement relying on IP/transport fields (e.g., the 5-tuple). IP-address-based policies may not be flexible enough to accommodate endpoints with volatile IP addresses.
- With the massive adoption of teleworking, there is a need to apply different security policies to the same set of endpoints under different circumstances (e.g., prevent relay attacks against a local attachment point to the enterprise network). For example, network access might be granted based upon criteria such as a user's access location, source network reputation, a user's role, the time of day, the type of network device used (e.g., corporate-issued device versus personal device), a device's security posture, etc. This means that the network needs to recognize the endpoints' identities and their current contexts and map the endpoints to their correct access grants to the network.

This document defines a YANG data model ([Section 5.2](#)) for policy-based network access control, which extends the IETF Access Control Lists (ACLs) module defined in [\[RFC8519\]](#). This module can be used to ensure consistent enforcement of ACL policies based on the group identity. Additionally, the YANG data model defined in the document also extends ACLs with date and time parameters to support schedule-aware policy enforcement.

The ACL concept has been generalized to be device-nonspecific, and it can be defined at the network/administrative domain level [\[RFC9899\]](#). To allow for all ACL applications, the YANG module for policy-based network ACL defined in [Section 5.2](#) does not limit how it can be used.

Specifically in scenarios where network access is triggered by user authentication, this document also defines a mechanism to establish a mapping between (1) the user group identifier (ID) and (2) common IP packet header fields and other encapsulating packet data (e.g., a Media Access Control (MAC) address) to execute the policy-based access control. Additionally, the document defines a Remote Authentication Dial-in User Service (RADIUS) [\[RFC2865\]](#) attribute that is used to communicate the user group identifier as part of identification and authorization information ([Section 6](#)).

Although this document cites MAC addresses as an example in some sections, this document does not make assumptions about which identifiers are used to trigger ACLs. These examples should not be considered as recommendations. Readers should be aware that MAC-based ACLs can be bypassed by clearing the MAC address. Other implications related to the change of MAC addresses are discussed in [\[RFC9797\]](#).

This document does not specify how to map the policy group identifiers to dedicated packet fields. Group-Based Policy (GBP), discussed in [Section 6.2.3](#) of [\[RFC9638\]](#), provides an example of how that may be achieved.

2. Conventions and Definitions

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [\[RFC2119\]](#) [\[RFC8174\]](#) when, and only when, they appear in all capitals, as shown here.

The meanings of the symbols in tree diagrams are defined in [\[RFC8340\]](#).

This document uses the following terms defined in [\[RFC8519\]](#):

- Access Control Entry (ACE)
- Access Control List (ACL)

The following definitions are used throughout this document:

Enterprise device: A device that falls under the access control domain of a centrally managed authority (enterprise administrator, typically). An enterprise device provides compute, memory, storage, and networking capabilities and connects to a network.

An enterprise device could be a server that hosts applications or software that delivers services to enterprise users. It could also be an enterprise Internet of Things (IoT) device that serves a limited purpose (e.g., a printer that allows users to scan and print).

While a personal device (BYOD) is not a physical asset of the enterprise, it is subject to the enterprise's access control policies when accessing the enterprise resources controlled by the centrally managed authority.

Endpoint: An entity that could be an end user, enterprise device, or application that actually connects to a network.

Endpoint group: A group of endpoints that share common access control policies.

User group: A group of end users who will be assigned the same network access policy. An end user is defined as a person. Refer to [Section 4.2.1](#) for more details.

Device group: A collection of enterprise devices that share common access control policies. Refer to [Section 4.2.2](#) for more details.

Application group: A collection of applications that share common access control policies. An application is a software program used for a specific service. Refer to [Section 4.2.3](#) for more details.

Endpoint group identifier: An identifier used to represent the collective identity of an endpoint group. An endpoint group may include a user group, device group, or application group.

User-group-based Control List (UCL) data model: A YANG data model for policy-based network access control that specifies an extension to the "ietf-access-control-list" module [[RFC8519](#)]. It allows policy enforcement based on a group identifier, which can be used both at the network device level and at the network/administrative domain level.

Policy: A set of rules to administer, manage, and control access to network resources [[RFC3198](#)].

3. Sample Usage

Access to some networks (e.g., enterprise networks) requires recognizing the endpoints' identities no matter how, where, or when they connect to the network resources. Then, the network maps the (connecting) endpoints to their access authorization rights. Such rights are defined using local policies. As discussed in [Section 1](#), because (1) there is a large number of connecting endpoints and (2) an endpoint may have different source IP addresses in different network segments, deploying a network access control policy for each IP address or network segment requires a high overhead. An alternate approach is to configure endpoint groups to classify users, enterprise devices, and applications, and to associate ACLs with endpoint groups so that endpoints in each group can share a group of ACL rules. This approach greatly reduces the overhead of the administrators and optimizes ACL resources.

The network ACLs can be provisioned on devices using specific mechanisms, such as those described in [[RFC8519](#)] or [[RFC9899](#)].

Different policies may need to be applied in different contextual situations. For example, companies may restrict (or grant) employees access to specific internal or external resources during work hours, while another policy is adopted during off-hours and weekends. A network administrator may also require traffic shaping ([Section 2.3.3.3 of \[RFC2475\]](#)) and policing ([Section 2.3.3.4 of \[RFC2475\]](#)) during peak hours in order to not affect other data services.

4. Policy-Based Network Access Control

4.1. Overview

An example architecture of a system that provides real-time and consistent enforcement of access control policies is shown in [Figure 1](#). This architecture illustrates a user-centric flow, which includes the following functional entities and interfaces:

- A service orchestrator that coordinates the overall service, including security policies. The service may be connectivity or any other access to resources that can be hosted and offered by a network.
- A Software-Defined Networking (SDN) [\[RFC7149\]](#) [\[RFC7426\]](#) controller that is responsible for maintaining endpoint-group-based ACLs and mapping the endpoint group to the associated attributes information (e.g., packet header fields). An SDN controller also behaves as a Policy Decision Point (PDP) [\[RFC3198\]](#) and pushes the required access control policies to relevant Policy Enforcement Points (PEPs) [\[RFC3198\]](#). A PDP is also known as a "policy server" [\[RFC2753\]](#).

An SDN controller may interact with an Authentication, Authorization, and Accounting (AAA) [\[RFC3539\]](#) server or a Network Access Server (NAS) [\[RFC7542\]](#).

- A NAS entity that handles authentication requests. The NAS interacts with a AAA server to complete user authentication using protocols like RADIUS [\[RFC2865\]](#). When access is granted, the AAA server provides the group identifier (group ID) to which the user belongs when the user first logs onto the network.

A new RADIUS attribute is defined in [Section 6](#) for this purpose.

- The AAA server provides a collection of authentication, authorization, and accounting functions. The AAA server is responsible for centralized user information management. The AAA server is preconfigured with user credentials (e.g., username and password), possible group identities, and related user attributes (users may be divided into different groups based on different user attributes).
- A PEP is the central entity that is responsible for enforcing appropriate access control policies. A first deployment scenario assumes that the SDN controller maps the group ID to the related common packet header and delivers ACL policies based on packet header fields to the required PEPs. Another deployment scenario may require that PEPs map incoming packets to their associated source and/or destination endpoint group IDs and act upon the corresponding group-based ACL policies (e.g., a group identifier may be carried in packet headers, as discussed in [Section 6.2.3 of \[RFC9638\]](#)).

Multiple PEPs may be involved in a network.

A PEP exposes a YANG-based interface (e.g., NETCONF [RFC6241]) to an SDN controller.

Figure 1 provides the overall architecture and procedure for policy-based access control management.

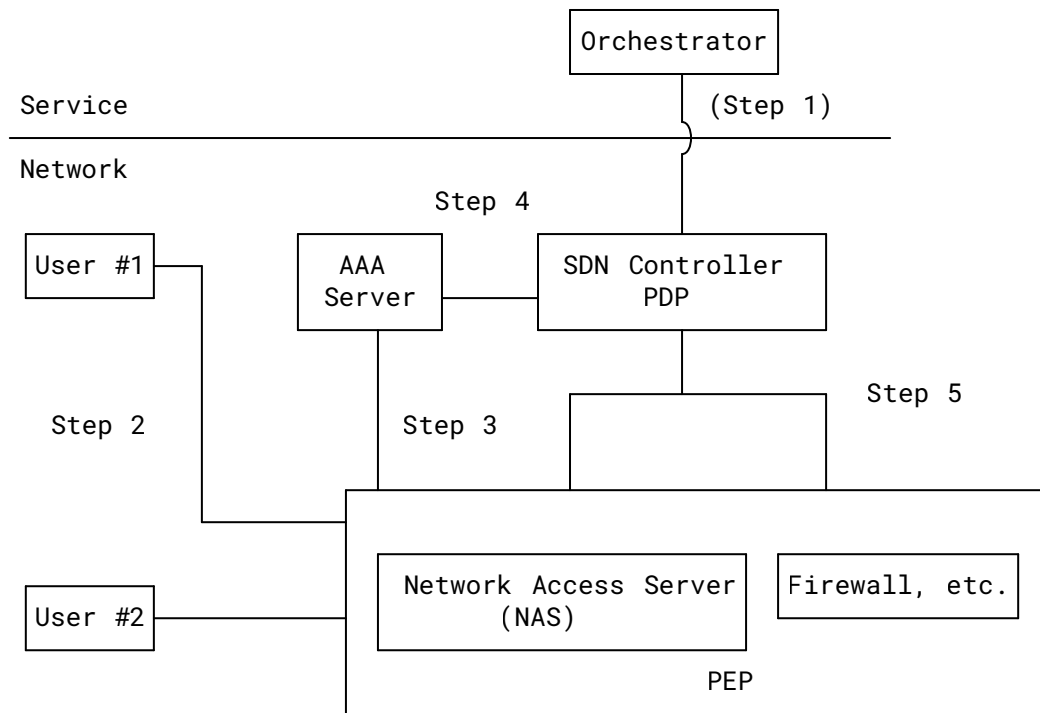


Figure 1: An Example Architecture for User-Group-Based Policy Management

In reference to Figure 1, the following typical flow is experienced:

- Step 1: Administrators (or a service orchestrator) configure an SDN controller with network-level ACLs using the YANG module defined in Section 5.2. An example is provided in Appendix A.1.
- Step 2: When a user first logs onto the network, they are required to be authenticated (e.g., using a username and password) at the NAS.
- Step 3: The authentication request is then relayed to the AAA server using a protocol such as RADIUS [RFC2865]. It is assumed that the AAA server has been appropriately configured to store user credentials, e.g., username, password, group information, and other user attributes. This document does not restrict what authentication method is used. Administrators may refer to, e.g., Section 7.4 of [RADIUS-DEPRECATE] for authentication method recommendations.

If the authentication request succeeds, the user is placed in a user group with the identifier returned to the NAS as the authentication result (see [Section 6](#)). If the authentication fails, the user is not assigned any user group, which also means that the user has no access (i.e., an Access-Reject is returned) or the user is assigned a special group with very limited access permissions for the network (as a function of the local policy). ACLs are enforced so that flows from the user's IP address are discarded (or rate-limited) by the network.

In some implementations, the AAA server can be integrated with the SDN controller.

Step 4: Either the AAA server or the NAS notifies the SDN controller of the mapping between the user group ID and related common packet header attributes (e.g., the 5-tuple). The exact details of how such notification is performed are out of scope of this specification.

Step 5: Either group-based access control policies or access control policies based on packet header fields are maintained on relevant PEPs under the SDN controller's management. Both types of ACL policy may exist on the PEP. Appendices [A.2](#) and [A.3](#) elaborate on each case.

A similar flow applies to policy management based on other endpoint group types, such as device or application groups, except that the mapping between the group ID and related common packet header attributes (e.g., 5-tuple) may be maintained on the SDN controller based on an inventory or an application registry. Particularly, the use of RADIUS exchanges is not required in such cases ([Section 6](#)).

[Section 8](#) provides additional operational considerations.

4.2. Endpoint Group

4.2.1. User Group

A user group is determined by a set of predefined policy criteria (e.g., source IP address, geolocation data, time of day, or device certificate). It uses an identifier (user group ID) to represent the collective identity of a group of users. Users may be moved to different user groups if there is a change in their composite attributes, environment, and/or local enterprise policy.

A user is authenticated, classified at the AAA server, and assigned to a user group. A user's group membership may change as aspects of the user change. For example, if the user group membership is determined solely by the source IP address, then a given user's group ID will change when the user is assigned a new IP address that falls outside of the range of addresses of the previous user group.

This document does not make any assumption about how user groups are defined. Such considerations are deployment-specific and are out of scope. However, and for illustration purposes, [Table 1](#) shows an example of how user group definitions may be characterized. User groups may share several common criteria. That is, user group criteria are not mutually exclusive. For example, the policy criteria of the user groups R&D Regular and R&D BYOD may share the same set of users that belong to the R&D organization but differ only in the type of

clients (corporate-issued clients vs. users' personal clients). Likewise, the same user may be assigned to different user groups depending on the time of day or the type of day (e.g., weekdays versus weekends), etc.

Group Name	Group ID	Group Description
R&D Regular	foo-10	R&D employees
R&D BYOD	foo-11	Personal devices of R&D employees
Sales	foo-20	Sales employees
VIP	foo-30	VIP employees

Table 1: User Group Examples

4.2.2. Device Group

A device group ID is an identifier that represents the collective identity of a group of enterprise devices. [Table 2](#) shows an example of how device group definitions may be characterized.

Group Name	Group ID	Group Description
Workflow	bar-40	Workflow resource servers
R&D Resource	bar-50	R&D resource servers
Printer Resource	bar-60	Printer resources

Table 2: Device Group Examples

Matching abstract device group IDs instead of specified addresses in ACL policies helps shield the consequences of address changes (e.g., back-end VM-based server migration).

4.2.3. Application Group

An application group is a collection of applications that share common access control policies. A device may run multiple applications, and different policies might need to be applied to the applications and device. A single application may need to run on multiple devices/VMs/containers; the abstraction of an application group eases the process of application migration. For example, the policy does not depend on the transport coordinates (i.e., 5-tuple). [Table 3](#) shows an example of how application group definitions may be characterized.

Group Name	Group ID	Group Description
Audio/Video Streaming	baz-70	Audio/Video conferencing application
Instant Messaging	baz-80	Messaging application
Document Collaboration	baz-90	Real-time document editing application

Table 3: Application Group Examples

4.3. Relations Between Different Endpoint Groups

Policy enforcement can be targeted to different endpoint groups in different scenarios. For example, when a user connects to the network and accesses an application hosted on one or multiple devices, access policies may be applied to different user groups. In some cases, applications and devices may operate and run without requiring any user interventions, or they may require user authentication, but access rules do not differentiate between different users. This enables policies to be applied to the application or device group. A device group can be used when there is only one single application running on the device or different applications running but with the same access control rules. If there is an application running on different devices/VMs/containers, it is simpler to apply a single policy to the application group.

5. The UCL Extension to the ACL Module

5.1. Module Overview

This module specifies an extension to the "ietf-access-control-list" module [RFC8519]. This extension adds endpoint groups so that an endpoint group identifier can be matched upon, and it also enables access control policy activation based on date and time conditions.

[Figure 2](#) provides the tree structure of the "ietf-ucl-acl" module.

```

module: ietf-ucl-acl

augment /acl:acls:
  +--rw endpoint-groups {ucl:group}?
  +--rw endpoint-group* [group-id]
  +--rw group-id      string
  +--rw group-type?   identityref
augment /acl:acls/acl:acl/acl:aces/acl:ace/acl:matches:
  +--rw endpoint-group {ucl:match-on-group}?
  +--rw source-group-id?      group-id-reference
  +--rw destination-group-id? group-id-reference
augment /acl:acls/acl:acl/acl:aces/acl:ace:
  +--rw effective-schedule {ucl:schedule}?
  +--rw (schedule-type)?
  +--:(period)
  |   +--rw period
  |   |   +--rw period-description?      string
  |   |   +--rw period-start?            yang:date-and-time
  |   |   +--rw time-zone-identifier?    sys:timezone-name
  |   |   +--rw (period-type)?
  |   |   |   +--:(explicit)
  |   |   |   |   +--rw period-end?      yang:date-and-time
  |   |   |   +--:(duration)
  |   |   |   |   +--rw duration?        duration
  |   +--:(recurrence)
  |   |   +--rw recurrence {schedule:icalendar-recurrence}?
  |   |   |   +--rw recurrence-first
  |   |   |   |   +--rw start-time?    yang:date-and-time
  |   |   |   |   +--rw duration?      duration
  |   |   |   +--rw time-zone-identifier? sys:timezone-name
  |   |   |   +--rw (recurrence-end)?
  |   |   |   |   +--:(until)
  |   |   |   |   |   +--rw until?      yang:date-and-time
  |   |   |   |   +--:(count)
  |   |   |   |   |   +--rw count?      uint32
  |   |   |   +--rw recurrence-description? string
  |   |   |   +--rw frequency?            identityref
  |   |   |   +--rw interval?             uint32
  |   |   |   +--rw period* [period-start]
  |   |   |   |   +--rw period-description?      string
  |   |   |   |   +--rw period-start            yang:date-and-time
  |   |   |   |   +--rw time-zone-identifier?    sys:timezone-name
  |   |   |   |   +--rw (period-type)?
  |   |   |   |   |   +--:(explicit)
  |   |   |   |   |   |   +--rw period-end?      yang:date-and-time
  |   |   |   |   |   +--:(duration)
  |   |   |   |   |   |   +--rw duration?        duration
  |   |   |   +--rw bysecond*                uint32
  |   |   |   +--rw byminute*                uint32
  |   |   |   +--rw byhour*                  uint32
  |   |   |   +--rw byday* [weekday]
  |   |   |   |   +--rw direction*    int32
  |   |   |   |   +--rw weekday      schedule:weekday
  |   |   |   +--rw bymonthday*          int32
  |   |   |   +--rw byyearday*           int32
  |   |   |   +--rw byyearweek*         int32
  |   |   |   +--rw byyearmonth*        uint32

```

+-rw bysetpos*	int32
+-rw workweek-start?	schedule:weekday
+-rw exception-dates*	yang:date-and-time

Figure 2: Tree Structure of the "ietf-uc1-acl" Module

The first part of the "ietf-uc1-acl" module augments the "acls" container in the "ietf-access-control-list" module [RFC8519] with an "endpoint-groups" container that includes an "endpoint-group" list, where each entry has a "group-id" that uniquely identifies the endpoint group and a "group-type" parameter to specify the endpoint group type.

"group-id" is defined as a string rather than an unsigned integer (e.g., uint32) to accommodate deployments that require some identification hierarchy within a domain. Such a hierarchy is meant to ease coordination within an administrative domain. There might be cases where a domain needs to tag packets with the group they belong to. The tagging does not need to mirror exactly the "group ID" used to populate the policy. How the "group-id" string is mapped to the tagging or field in the packet header in an encapsulation scenario is outside the scope of this document. Augmentation may be considered in the future to cover encapsulation considerations.

The second part of the "ietf-uc1-acl" module augments the "matches" container in the "ietf-access-control-list" module [RFC8519] so that a source and/or destination endpoint group ID can be referenced as the match criteria.

The third part of the module augments the "ace" list in the "ietf-access-control-list" module [RFC8519] with date- and time-specific parameters to allow an ACE to be activated based on a date/time condition. Two types of time ranges ("period" and "recurrence") are defined, which reuse the "period-of-time" and "icalendar-recurrence" groupings, respectively, defined in the "ietf-schedule" YANG module [RFC9922].

5.2. The "ietf-uc1-acl" YANG Module

This module imports types and groupings defined in the "ietf-schedule" module [RFC9922]. It also augments the "ietf-access-control-list" module (Section 4.1 of [RFC8519]).

```
<CODE BEGINS> file "ietf-uc1-acl@2026-10-07.yang"

module ietf-uc1-acl {
  yang-version 1.1;
  namespace "urn:ietf:params:xml:ns:yang:ietf-uc1-acl";
  prefix uc1;

  import ietf-access-control-list {
    prefix acl;
    reference
      "RFC 8519: YANG Data Model for Network Access
      Control Lists (ACLs)";
  }
  import ietf-schedule {
    prefix schedule;
  }
}
```

```
reference
  "RFC 9922: A Common YANG Data Model for Scheduling";
}

organization
  "IETF OPSAWG (Operations and Management Area Working Group)";
contact
  "WG Web:  https://datatracker.ietf.org/wg/opsawg
  WG List:  OPSAWG <mailto:opsawg@ietf.org>

  Editor:    Qiufang Ma
             <mailto:maqiufang1@huawei.com>
  Author:    Qin Wu
             <mailto:bill.wu@huawei.com>
  Editor:    Mohamed Boucadair
             <mailto:mohamed.boucadair@orange.com>
  Author:    Daniel King
             <mailto:d.king@lancaster.ac.uk>;

description
  "The User-group-based Control List (UCL) YANG module augments
  the IETF Access Control Lists (ACLs) module.  UCL is meant
  to ensure consistent enforcement of ACL policies based on
  the group identity.

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  BSD License set forth in Section 4.c of the IETF Trust's
  Legal Provisions Relating to IETF Documents
  (https://trustee.ietf.org/license-info).

  All revisions of IETF and IANA published modules can be found
  at the YANG Parameters registry group
  (https://www.iana.org/assignments/yang-parameters).

  This version of this YANG module is part of RFC 10065; see
  the RFC itself for full legal notices.";

revision 2026-10-07 {
  description
    "Initial revision.";
  reference
    "RFC 10065: A YANG Data Model and RADIUS Extension for
    Policy-Based Network Access Control";
}

feature schedule {
  description
    "Indicates support of schedule-based Access Control
    Entries (ACEs).";
}

feature match-on-group {
  description
    "Indicates support of matching on endpoint groups.";
```

```
}

feature group {
  if-feature "ucl:match-on-group";
  description
    "Indicates support of group-based ACLs.";
}

feature mixed-ipv4-group {
  if-feature "acl:match-on-ipv4 and ucl:match-on-group";
  description
    "IPv4 and group ACL combinations supported.";
}

feature mixed-ipv6-group {
  if-feature "acl:match-on-ipv6 and ucl:match-on-group";
  description
    "IPv6 and group ACL combinations supported.";
}

feature mixed-ipv4-ipv6-group {
  if-feature "acl:match-on-ipv4 and acl:match-on-ipv6 and "
    + "ucl:match-on-group";
  description
    "IPv4, IPv6, and group ACL combinations supported.";
}

feature mixed-eth-group {
  if-feature "acl:match-on-eth and ucl:match-on-group";
  description
    "Ethernet and group ACL combinations supported.";
}

feature mixed-eth-ipv4-group {
  if-feature "acl:match-on-eth and acl:match-on-ipv4 and "
    + "ucl:match-on-group";
  description
    "Ethernet, IPv4, and group ACL combinations supported.";
}

feature mixed-eth-ipv6-group {
  if-feature "acl:match-on-eth and acl:match-on-ipv6 and "
    + "ucl:match-on-group";
  description
    "Ethernet, IPv6, and group ACL combinations supported.";
}

feature mixed-eth-ipv4-ipv6-group {
  if-feature "acl:match-on-eth and acl:match-on-ipv4 and "
    + "acl:match-on-ipv6 and ucl:match-on-group";
  description
    "Ethernet, IPv4, IPv6, and group ACL combinations supported.";
}

identity group-acl-type {
  if-feature "group";
  base acl:acl-base;
  description
```

```
"An ACL that matches based on an endpoint group identifier,
which can represent the collective identity of a group of
authenticated users, end devices, or applications. An
endpoint group identifier may be carried in the outer/inner
packet header (e.g., via Network Virtualization over Layer 3
(NV03) encapsulation) or may not correspond to any field in
the packet header. Matching on Layer 4 header fields may
also exist in the ACEs.";
}

identity mixed-ipv4-group-type {
  if-feature "mixed-ipv4-group";
  base acl:ipv4-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the IPv4 header and endpoint group identifiers, which can
    represent the collective identity of a group of authenticated
    users, end devices, or applications. Matching on Layer 4
    header fields may also exist in the ACEs.";
}

identity mixed-ipv6-group-type {
  if-feature "mixed-ipv6-group";
  base acl:ipv6-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the IPv6 header and endpoint group identifiers, which can
    represent the collective identity of a group of authenticated
    users, end devices, or applications. Matching on Layer 4
    header fields may also exist in the ACEs.";
}

identity mixed-ipv4-ipv6-group-type {
  if-feature "mixed-ipv4-ipv6-group";
  base acl:ipv4-acl-type;
  base acl:ipv6-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the IPv4 header, IPv6 header, and endpoint group
    identifiers, which can represent the collective identity of
    a group of authenticated users, end devices, or applications.
    Matching on Layer 4 header fields may also exist in the
    ACEs.";
}

identity mixed-eth-group-type {
  if-feature "mixed-eth-group";
  base acl:eth-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the Ethernet header and endpoint group identifiers,
    which can represent the collective identity of a group of
    authenticated users, end devices, or applications. Matching
    on Layer 4 header fields may also exist in the ACEs.";
```

```
}

identity mixed-eth-ipv4-group-type {
  if-feature "mixed-eth-ipv4-group";
  base acl:eth-acl-type;
  base acl:ipv4-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the Ethernet header, IPv4 header, and endpoint group
    identifiers, which can represent the collective identity of
    a group of authenticated users, end devices, or applications.
    Matching on Layer 4 header fields may also exist in the
    ACEs.";
}

identity mixed-eth-ipv6-group-type {
  if-feature "mixed-eth-ipv6-group";
  base acl:eth-acl-type;
  base acl:ipv6-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the Ethernet header, IPv6 header, and endpoint group
    identifiers, which can represent the collective identity of
    a group of authenticated users, end devices, or applications.
    Matching on Layer 4 header fields may also exist in the
    ACEs.";
}

identity mixed-eth-ipv4-ipv6-group-type {
  if-feature "mixed-eth-ipv4-ipv6-group";
  base acl:eth-acl-type;
  base acl:ipv4-acl-type;
  base acl:ipv6-acl-type;
  base ucl:group-acl-type;
  description
    "An ACL that contains a mix of entries that match on fields
    in the Ethernet header, IPv4 header, IPv6 header, and
    endpoint group identifiers, which can represent the collective
    identity of a group of authenticated users, end devices, or
    applications. Matching on Layer 4 header fields may also
    exist in the ACEs.";
}

identity endpoint-group-type {
  description
    "Identity for the type of endpoint group.";
}

identity user-group {
  base ucl:endpoint-group-type;
  description
    "Indicates user endpoint group type.";
}

identity device-group {
  base ucl:endpoint-group-type;
```

```

    description
      "Indicates device endpoint group type.";
  }

  identity application-group {
    base ucl:endpoint-group-type;
    description
      "Indicates application endpoint group type.";
  }

  typedef group-id-reference {
    type leafref {
      path "/acl:acls/ucl:endpoint-groups"
        + "/ucl:endpoint-group/ucl:group-id";
    }
    description
      "Defines a reference to a group identifier.";
  }

  augment "/acl:acls" {
    if-feature "ucl:group";
    description
      "Adds a container for endpoint group definition.";
    container endpoint-groups {
      description
        "Defines a container for the endpoint group list.";
      list endpoint-group {
        key "group-id";
        description
          "Definition of the endpoint group list.";
        leaf group-id {
          type string {
            length "1..64";
          }
          description
            "The endpoint group identifier that uniquely identifies
              an endpoint group.";
        }
        leaf group-type {
          type identityref {
            base endpoint-group-type;
          }
          description
            "Specifies the type of the endpoint group (e.g., user,
              device, or application).  When not configured, the
              group is considered a generic or untyped endpoint
              group.";
        }
      }
    }
  }

  augment "/acl:acls/acl:acl/acl:aces/acl:ace/acl:matches" {
    if-feature "ucl:match-on-group";
    description
      "Specifies how a source and/or destination endpoint group
        ID can be referenced as the match criteria in the ACEs.";
    container endpoint-group {

```

```

when "derived-from-or-self(/acl:acls/acl:acl/acl:type, "
+ "'ucl:group-acl-type')";
description
  "Adds new match criteria based on the group identifier
  associated with the packet's source and/or
  destination endpoint.

  Note that this container is only valid when the ACL
  type is equal to or derived from 'group-acl-type',
  which depends on the 'group' feature. That is,
  implementations advertising 'match-on-group' need to
  also support the 'group' feature to ensure the
  validity of this container.";
leaf source-group-id {
  type group-id-reference;
  description
    "The matched source endpoint group identifier.";
}
leaf destination-group-id {
  type group-id-reference;
  description
    "The matched destination endpoint group identifier.";
}
}
}

augment "/acl:acls/acl:acl/acl:aces/acl:ace" {
  if-feature "ucl:schedule";
  description
    "Adds schedule parameters to allow the ACE to take effect
    based on date and time.";
  container effective-schedule {
    description
      "Defines when the access control entry rules
      are applied based on date and time conditions.
      If it is not configured, the ACE is immediately
      and always applied.";
    choice schedule-type {
      description
        "Choice based on the type of the time range.";
      container period {
        description
          "The ACE is applied based on a precise period of
          time.";
        uses schedule:period-of-time;
      }
      container recurrence {
        if-feature "schedule:icalendar-recurrence";
        description
          "The ACE is applied based on a recurrence rule.";
        uses schedule:icalendar-recurrence;
      }
    }
  }
}
}
}
}
<CODE ENDS>

```

6. User-Access-Group-ID RADIUS Attribute

This section defines the User-Access-Group-ID RADIUS attribute, which is designed for user-centric access control scenarios where network access is triggered by user authentication and used to indicate the user group ID to be used by the NAS. For other endpoint group types, such as device or application groups, the identifiers are typically preprovisioned on the SDN controller based on an inventory or an application registry.

The definition of the attribute follows the guidelines in [Section 2.7.1](#) of [\[RFC6929\]](#). When the User-Access-Group-ID RADIUS attribute is present in the RADIUS Access-Accept, the system applies the related access control to the user after the user authenticates.

The User-Access-Group-ID RADIUS attribute is of type "string" as defined in [Section 3.5](#) of [\[RFC8044\]](#).

The User-Access-Group-ID RADIUS attribute **MAY** appear in a RADIUS Access-Accept packet. It **MAY** also appear in a RADIUS Access-Request packet as a hint to the RADIUS server to indicate a preference. However, the server is not required to honor such a preference. If more than one instance of the User-Access-Group-ID RADIUS attribute appears in a RADIUS Access-Accept packet, it means that the user is a member of many groups.

The User-Access-Group-ID RADIUS attribute **MAY** appear in a RADIUS CoA-Request packet.

The User-Access-Group-ID RADIUS attribute **MAY** appear in a RADIUS Accounting-Request packet. Specifically, this may be used by a NAS to acknowledge that the attribute was received in the RADIUS Access-Accept and the NAS is enforcing that policy.

The User-Access-Group-ID RADIUS attribute **MUST NOT** appear in any other RADIUS packet.

The User-Access-Group-ID RADIUS attribute is structured as follows:

Type: 241.12

Length: This field indicates the total length, in octets, of all fields of this attribute, including the Type, Length, Extended-Type, and Value. The Length **MUST** at least 4 octets and **MUST NOT** be more than 67 octets. The maximum length is 67 octets to accommodate the maximum group ID of 64 octets, plus one octet each for Type, Length, and Extended-Type.

Data Type: string ([Section 3.5](#) of [\[RFC8044\]](#)).

Value: This field contains the user group ID.

7. Table of RADIUS Attributes

[Table 4](#) provides a guide that indicates which types of RADIUS packets may contain a User-Access-Group-ID RADIUS attribute and in what quantity.

Access-Request	Access-Accept	Access-Reject	Access-Challenge	Attribute
0+	0+	0	0	User-Access-Group-ID
Accounting-Request	CoA-Request	CoA-ACK	CoA-NACK	Attribute
0+	0+	0	0	User-Access-Group-ID

Table 4: Table of Attributes

Notation for Table 4:

0 This attribute **MUST NOT** be present in the packet.

0+ Zero or more instances of this attribute **MAY** be present in the packet.

8. Operational Considerations

8.1. Deployment Options

The UCL data model can be implemented in different ways.

In some cases, the UCL data model is implemented at the network/administrative domain level, where an SDN controller maintains the dynamic mapping from an endpoint group ID to IP/transport fields (e.g., the 5-tuple) and programs the PEPs with IP-address-based or 5-tuple-based ACLs. In such cases, PEPs do not require implementing specific logic (including hardware) compared to the enforcement of conventional ACLs.

It is possible for the UCL data model to be implemented at the device level. While it eliminates the need for an SDN controller to interact frequently with the PEPs for reasons like the user's context of network connection change or VM/application migration, dedicated hardware/software support might be needed for PEPs to understand the endpoint group identifier. In scenarios where the NAS behaves as the PEP that acquires the source and/or destination endpoint group ID from the AAA server, ACL policy enforcement based on the group ID without being encapsulated into packet headers might affect the forwarding performance. Implementations need to evaluate the operational trade-off (flexibility brought to the network vs. the complexity of implementation) carefully. Such an assessment is out of scope for this document.

8.2. Hardware/Software Implications

Some devices may not have built-in capabilities to enforce group-based match policies. Hardware or software upgrades may be required to support such features by the involved PEPs.

8.3. Mapping Consistency

This specification requires that an adequate setup is put in place to map a group ID to packet fields, typically managed by a controller. Special care should be taken to ensure that such mapping is appropriately enforced when distinct mechanisms (RADIUS, etc.) are supported in the network.

9. Security Considerations

9.1. YANG

This section is modeled after the template described in [Section 3.7.1](#) of [\[RFC9907\]](#).

The "ietf-ucl-acl" YANG module defines a data model that is designed to be accessed via YANG-based management protocols, such as the Network Configuration Protocol (NETCONF) [\[RFC6241\]](#) and RESTCONF [\[RFC8040\]](#). These YANG-based management protocols (1) have to use a secure transport layer (e.g., Secure Shell (SSH) [\[RFC4252\]](#), TLS [\[RFC9846\]](#), and QUIC [\[RFC9000\]](#)) and (2) have to use mutual authentication.

The Network Configuration Access Control Model (NACM) [\[RFC8341\]](#) provides the means to restrict access for particular NETCONF or RESTCONF users to a preconfigured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data nodes defined in this YANG module that are writable/creatable/deletable (i.e., "config true", which is the default). All writable data nodes are likely to be sensitive or vulnerable in some network environments. Write operations (e.g., edit-config) and delete operations to these data nodes without proper protection or authentication can have a negative effect on network operations. The following subtrees and data nodes have particular sensitivities/vulnerabilities:

- /acl:acls/ucl:endpoint-groups/ucl:endpoint-group:

This list specifies all the endpoint group entries. Unauthorized write access to this list can allow intruders to modify the entries so as to forge an endpoint group that does not exist or maliciously delete an existing endpoint group, which could be used to craft an attack.

- /acl:acls/acl:acl/acl:aces/acl:ace/acl:matches/ucl:endpoint-group:

This subtree specifies a source and/or destination endpoint group ID as match criteria in the ACEs. Unauthorized write access to this data node may allow intruders to modify the group ID so as to permit access that should not be permitted, or deny access that should be permitted.

- /acl:acls/acl:acl/acl:aces/acl:ace/ucl:effective-schedule:

It specifies the scheduling of ACLs. Unauthorized write access to this data node may allow intruders to alter it. This may lead to service disruption or unavailability. Strict access control is needed for write operations on this subtree to ensure that only authorized users can modify it.

Some of the readable data nodes in this YANG module may be considered sensitive or vulnerable in some network environments. It is thus important to control read access (e.g., via get, get-config, or notification) to these data nodes. Specifically, the following subtrees and data nodes have particular sensitivities/ vulnerabilities:

- /acl:acls/acl:acl/acl:aces/acl:ace/ucl:effective-schedule:

It specifies when the access control entry rules are applied. Unauthorized read access of the list will allow an attacker to determine which rules are applied, to better craft an attack.

This YANG module uses groupings from other YANG modules that define nodes that may be considered sensitive or vulnerable in network environments. Refer to the Security Considerations of [RFC9922] for information as to which nodes may be considered sensitive or vulnerable in network environments.

9.2. RADIUS

RADIUS-related security considerations are discussed in [RFC2865]. An effort to deprecate insecure practices in RADIUS is provided in [RADIUS-DEPRECATE].

This document targets deployments where a trusted relationship is in place between the RADIUS client and server with communication optionally secured by IPsec or Transport Layer Security (TLS) [RFC6614] [RadSec].

10. IANA Considerations

10.1. YANG

IANA has assigned the following URI in the "ns" registry within the "IETF XML Registry" group [RFC3688]:

URI: urn:ietf:params:xml:ns:yang:ietf-ucl-acl

Registrant Contact: The IESG

XML: N/A; the requested URI is an XML namespace.

IANA has registered the following YANG module in the "YANG Module Names" registry [RFC6020] within the "YANG Parameters" registry group:

Name: ietf-ucl-acl

Maintained by IANA? N

Namespace: urn:ietf:params:xml:ns:yang:ietf-ucl-acl

Prefix: ucl

Reference: RFC 10065

10.2. RADIUS

IANA has assigned the following attribute type in the "RADIUS Attribute Types" registry within the "RADIUS Types" registry group [RADIUS-Types]:

Value	Description	Data Type	Reference
241.12	User-Access-Group-ID	string	RFC 10065

Table 5: RADIUS Attribute

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Appendix A. Usage Examples

A.1. Configuring the Controller Using the Group-Based ACL

Let's consider an organization that would like to manage the access of R&D employees who bring personally owned devices (BYOD) into the workplace.

The access requirements are as follows:

- Permit traffic from R&D employees' personal devices, destined to R&D employees' devices, every work day from 8:00:00 to 18:00:00 UTC, starting on January 1, 2026.
- Deny traffic from R&D employees' personal devices, destined to finance servers located in the enterprise data center (DC) network, starting at 8:30:00 on January 20, 2026, with an offset of -08:00 from UTC (Pacific Standard Time), and ending at 18:00:00 in Pacific Standard Time on December 31, 2026.

The example shown in [Figure 3](#) illustrates the configuration of an SDN controller using the group-based ACL:

```
{
  "ietf-access-control-list:acls": {
    "ietf-ucl-acl:endpoint-groups": {
      "endpoint-group": [
        {
          "group-id": "R&D",
          "group-type": "ietf-ucl-acl:user-group"
        },
        {
          "group-id": "R&D BYOD",
          "group-type": "ietf-ucl-acl:user-group"
        },
        {
          "group-id": "finance server",
          "group-type": "ietf-ucl-acl:device-group"
        }
      ]
    },
    "acl": [
      {
        "name": "sample-group-acl",
        "type": "ietf-ucl-acl:group-acl-type",
        "aces": {
          "ace": [
            {
              "name": "rule1",
              "matches": {
                "ietf-ucl-acl:endpoint-group": {
                  "source-group-id": "R&D BYOD",
                  "destination-group-id": "R&D"
                }
              },
              "actions": {
                "forwarding": "ietf-access-control-list:accept"
              },
              "ietf-ucl-acl:effective-schedule": {
                "recurrence": {
                  "recurrence-first": {
                    "start-time": "2026-01-01T08:00:00Z",
                    "duration": "PT10:00:00"
                  },
                  "frequency": "ietf-schedule:daily",
                  "byday": [
                    {
                      "weekday": "monday"
                    },
                    {
                      "weekday": "tuesday"
                    },
                    {
                      "weekday": "wednesday"
                    },
                    {
                      "weekday": "thursday"
                    },
                    {
                      "weekday": "friday"
                    }
                  ]
                }
              }
            }
          ]
        }
      }
    ]
  }
}
```

```
{
  "name": "rule2",
  "matches": {
    "ietf-uc1-acl:endpoint-group": {
      "source-group-id": "R&D BYOD",
      "destination-group-id": "finance server"
    }
  },
  "actions": {
    "forwarding": "ietf-access-control-list:reject"
  },
  "ietf-uc1-acl:effective-schedule": {
    "period": {
      "period-start": "2026-01-20T08:30:00-08:00",
      "period-end": "2026-12-31T18:00:00-08:00"
    }
  }
}
```

Figure 3: Example of UCL Configuration on the SDN Controller

A.2. Configuring a PEP Using the Group-Based ACL

This section illustrates an example of configuring a PEP using the group-based ACL.

The PEP that enforces a group-based ACL may acquire group IDs from the AAA server if working as a NAS authenticating both the source endpoint and destination endpoint users. Another case for a PEP enforcing a group-based ACL is to obtain the group ID of the source endpoint directly from a packet field [VXLAN].

Assume the mapping between a device group ID and IP addresses is predefined or acquired via device authentication. Figure 4 shows the ACL configuration delivered from the controller to the PEP. This example is consistent with the example presented in Appendix A.1.

The examples in this section do not intend to be exhaustive. In particular, explicit IP addresses ("destination-ipv4-network" or "destination-ipv6-network") are provided only for one single rule to illustrate how the mapping between a group ID and IP addresses is translated into an ACL rule entry.

```
{
  "ietf-access-control-list:acls": {
    "ietf-uc1-acl:endpoint-groups": {
      "endpoint-group": [
        {
          "group-id": "R&D",
          "group-type": "ietf-uc1-acl:user-group"
        },
        {
          "group-id": "R&D BYOD",
          "group-type": "ietf-uc1-acl:user-group"
        }
      ]
    },
    "acl": [
      {
        "name": "sample-uc1-ipv4",
        "type": "ietf-uc1-acl:mixed-ipv4-group-type",
        "aces": {
          "ace": [
            {
              "name": "rule1",
              "matches": {
                "ietf-uc1-acl:endpoint-group": {
                  "source-group-id": "R&D BYOD",
                  "destination-group-id": "R&D"
                }
              },
              "actions": {
                "forwarding": "ietf-access-control-list:accept"
              },
              "ietf-uc1-acl:effective-schedule": {
                "recurrence": {
                  "recurrence-first": {
                    "start-time": "2026-01-01T08:00:00Z",
                    "duration": "PT10:00:00"
                  },
                  "frequency": "ietf-schedule:daily",
                  "byday": [
                    {
                      "weekday": "monday"
                    },
                    {
                      "weekday": "tuesday"
                    },
                    {
                      "weekday": "wednesday"
                    },
                    {
                      "weekday": "thursday"
                    },
                    {
                      "weekday": "friday"
                    }
                  ]
                }
              }
            }
          ]
        }
      }
    ]
  }
}
```

```
    },  
    {  
      "name": "rule2",  
      "matches": {  
        "ietf-uc1-acl:endpoint-group": {  
          "source-group-id": "R&D BYOD"  
        },  
        "ipv4": {  
          "destination-ipv4-network": "203.0.113.1/24"  
        }  
      },  
      "actions": {  
        "forwarding": "ietf-access-control-list:reject"  
      },  
      "ietf-uc1-acl:effective-schedule": {  
        "period": {  
          "period-start": "2026-01-20T08:30:00-08:00",  
          "period-end": "2026-12-31T18:00:00-08:00"  
        }  
      }  
    }  
  ]  
}  
}
```

Figure 4: Example of PEP Configuration Using a Group-Based ACL

[Figure 5](#) shows an example of the same policy but with a destination IPv6 prefix.

```
{
  "ietf-access-control-list:acls": {
    "ietf-ucl-acl:endpoint-groups": {
      "endpoint-group": [
        {
          "group-id": "R&D",
          "group-type": "ietf-ucl-acl:user-group"
        },
        {
          "group-id": "R&D BYOD",
          "group-type": "ietf-ucl-acl:user-group"
        }
      ]
    },
    "acl": [
      {
        "name": "sample-ucl-ipv6",
        "type": "ietf-ucl-acl:mixed-ipv6-group-type",
        "aces": {
          "ace": [
            {
              "name": "rule1",
              "matches": {
                "ietf-ucl-acl:endpoint-group": {
                  "source-group-id": "R&D BYOD",
                  "destination-group-id": "R&D"
                }
              },
              "actions": {
                "forwarding": "ietf-access-control-list:accept"
              },
              "ietf-ucl-acl:effective-schedule": {
                "recurrence": {
                  "recurrence-first": {
                    "start-time": "2026-01-01T08:00:00Z",
                    "duration": "PT10:00:00"
                  },
                  "frequency": "ietf-schedule:daily",
                  "byday": [
                    {
                      "weekday": "monday"
                    },
                    {
                      "weekday": "tuesday"
                    },
                    {
                      "weekday": "wednesday"
                    },
                    {
                      "weekday": "thursday"
                    },
                    {
                      "weekday": "friday"
                    }
                  ]
                }
              }
            }
          ]
        }
      }
    ]
  }
}
```

```

    },
    {
      "name": "rule2",
      "matches": {
        "ietf-uc1-acl:endpoint-group": {
          "source-group-id": "R&D BYOD"
        },
        "ipv6": {
          "destination-ipv6-network": "2001:db8:1234::/64"
        }
      },
      "actions": {
        "forwarding": "ietf-access-control-list:reject"
      },
      "ietf-uc1-acl:effective-schedule": {
        "period": {
          "period-start": "2026-01-20T08:30:00-08:00",
          "period-end": "2026-12-31T18:00:00-08:00"
        }
      }
    }
  ]
}

```

Figure 5: Example of PEP Configuration Using a Group-Based ACL (IPv6)

A.3. Configuring a PEP Using an Address-Based ACL

This section describes an example of configuring a PEP using an IP-address-based ACL. IP-address-based access control policies could be applied to a PEP that may not understand the group information (e.g., a firewall).

Assume an employee in the R&D department accesses the network wirelessly from a non-corporate laptop. The SDN controller associates the user group to which the employee belongs with the user's address according to steps 1 to 4 in [Section 4.1](#).

Assume the mapping between a device group ID and IP addresses is predefined or acquired via device authentication. [Figure 6](#) shows an IPv4-address-based ACL configuration delivered from the controller to the PEP. This example is consistent with the example presented in [Appendix A.1](#).

```
{
  "ietf-access-control-list:acls": {
    "acl": [
      {
        "name": "sample-acl-ipv4",
        "type": "ietf-access-control-list:ipv4-acl-type",
        "aces": {
          "ace": [
            {
              "name": "rule1",
              "matches": {
                "ipv4": {
                  "destination-ipv4-network": "192.168.2.1/24",
                  "source-ipv4-network": "192.168.1.1/24"
                }
              },
              "actions": {
                "forwarding": "ietf-access-control-list:accept"
              },
              "ietf-uc1-acl:effective-schedule": {
                "recurrence": {
                  "recurrence-first": {
                    "start-time": "2026-01-01T08:00:00Z",
                    "duration": "PT10:00:00"
                  },
                  "frequency": "ietf-schedule:daily",
                  "byday": [
                    {
                      "weekday": "monday"
                    },
                    {
                      "weekday": "tuesday"
                    },
                    {
                      "weekday": "wednesday"
                    },
                    {
                      "weekday": "thursday"
                    },
                    {
                      "weekday": "friday"
                    }
                  ]
                }
              }
            }
          ]
        }
      },
      {
        "name": "rule2",
        "matches": {
          "ipv4": {
            "destination-ipv4-network": "203.0.113.1/24",
            "source-ipv4-network": "192.168.1.1/24"
          }
        },
        "actions": {
          "forwarding": "ietf-access-control-list:reject"
        }
      }
    ]
  }
}
```

```
    "ietf-uc1-acl:effective-schedule": {  
      "period": {  
        "period-start": "2026-01-20T08:30:00-08:00",  
        "period-end": "2026-12-31T18:00:00-08:00"  
      }  
    }  
  }  
} ]  
}  
}  
}
```

Figure 6: Example of PEP Configuration Using an Address-Based ACL

[Figure 7](#) shows an example of the same policy but with IPv6 prefixes.

```
{
  "ietf-access-control-list:acls": {
    "acl": [
      {
        "name": "sample-acl-ipv6",
        "type": "ietf-access-control-list:ipv6-acl-type",
        "aces": {
          "ace": [
            {
              "name": "rule1",
              "matches": {
                "ipv6": {
                  "destination-ipv6-network": "2001:db8:0:2::/64",
                  "source-ipv6-network": "2001:db8:0:1::/64"
                }
              },
              "actions": {
                "forwarding": "ietf-access-control-list:accept"
              },
              "ietf-uc1-acl:effective-schedule": {
                "recurrence": {
                  "recurrence-first": {
                    "start-time": "2026-01-01T08:00:00Z",
                    "duration": "PT10:00:00"
                  },
                  "frequency": "ietf-schedule:daily",
                  "byday": [
                    {
                      "weekday": "monday"
                    },
                    {
                      "weekday": "tuesday"
                    },
                    {
                      "weekday": "wednesday"
                    },
                    {
                      "weekday": "thursday"
                    },
                    {
                      "weekday": "friday"
                    }
                  ]
                }
              }
            }
          ]
        }
      },
      {
        "name": "rule2",
        "matches": {
          "ipv6": {
            "destination-ipv6-network": "2001:db8:1234::/64",
            "source-ipv6-network": "2001:db8:0:1::/64"
          }
        },
        "actions": {
          "forwarding": "ietf-access-control-list:reject"
        }
      }
    ]
  }
}
```

```
    "ietf-uc1-acl:effective-schedule": {  
      "period": {  
        "period-start": "2026-01-20T08:30:00-08:00",  
        "period-end": "2026-12-31T18:00:00-08:00"  
      }  
    }  
  }  
} ]  
}  
}  
]  
}  
}
```

Figure 7: Example of PEP Configuration Using an Address-Based ACL (IPv6)

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Authors' Addresses

Qiufang Ma (EDITOR)

Huawei
101 Software Avenue, Yuhua District
Jiangsu
210012
China
Email: maqiufang1@huawei.com

Qin Wu

Huawei

101 Software Avenue, Yuhua District

Jiangsu

210012

China

Email: bill.wu@huawei.com**Mohamed Boucadair (EDITOR)**

Orange

35000 Rennes

France

Email: mohamed.boucadair@orange.com**Daniel King**

Lancaster University

United Kingdom

Email: d.king@lancaster.ac.uk